What’s happening?

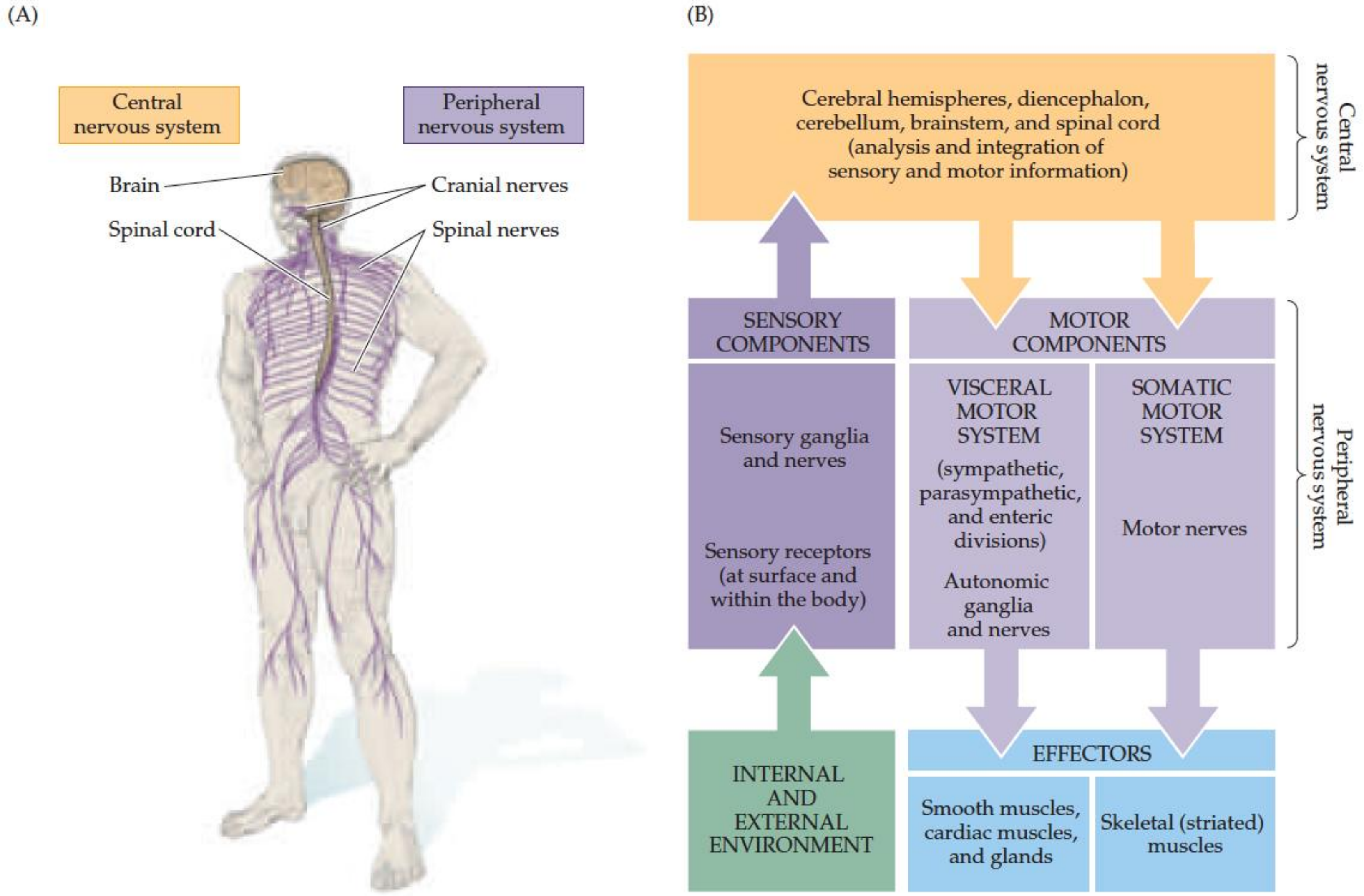
Electrical signals in neurons: the resting membrane.

MATTEO DIANO, PH.D.

COMPUTATIONAL APPLICATION TO COGNITIVE NEUROSCIENCE

UNIVERSITÀ DI TORINO

Our Protagonist

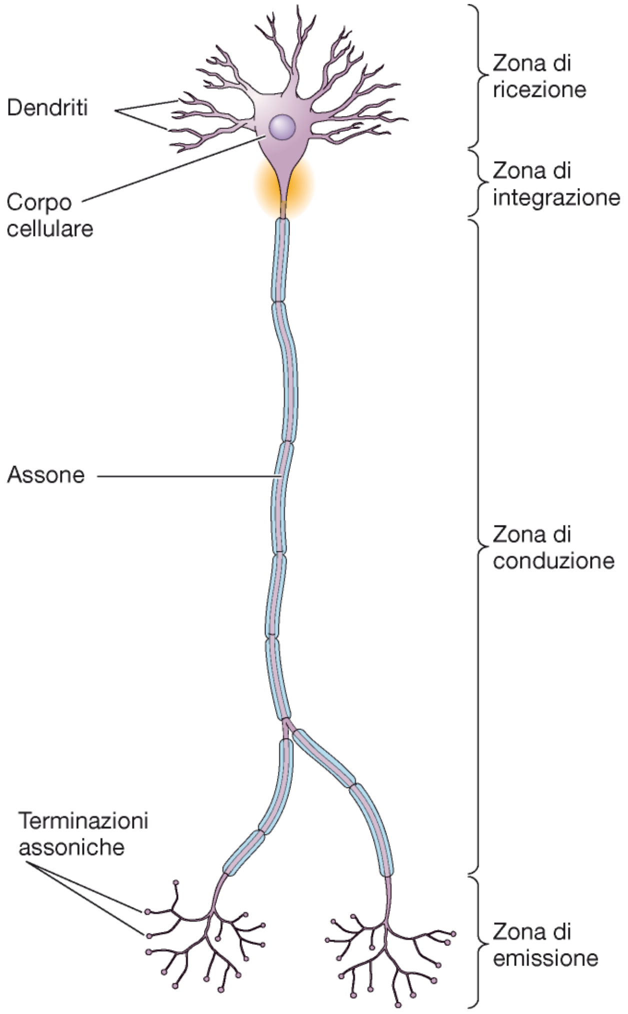


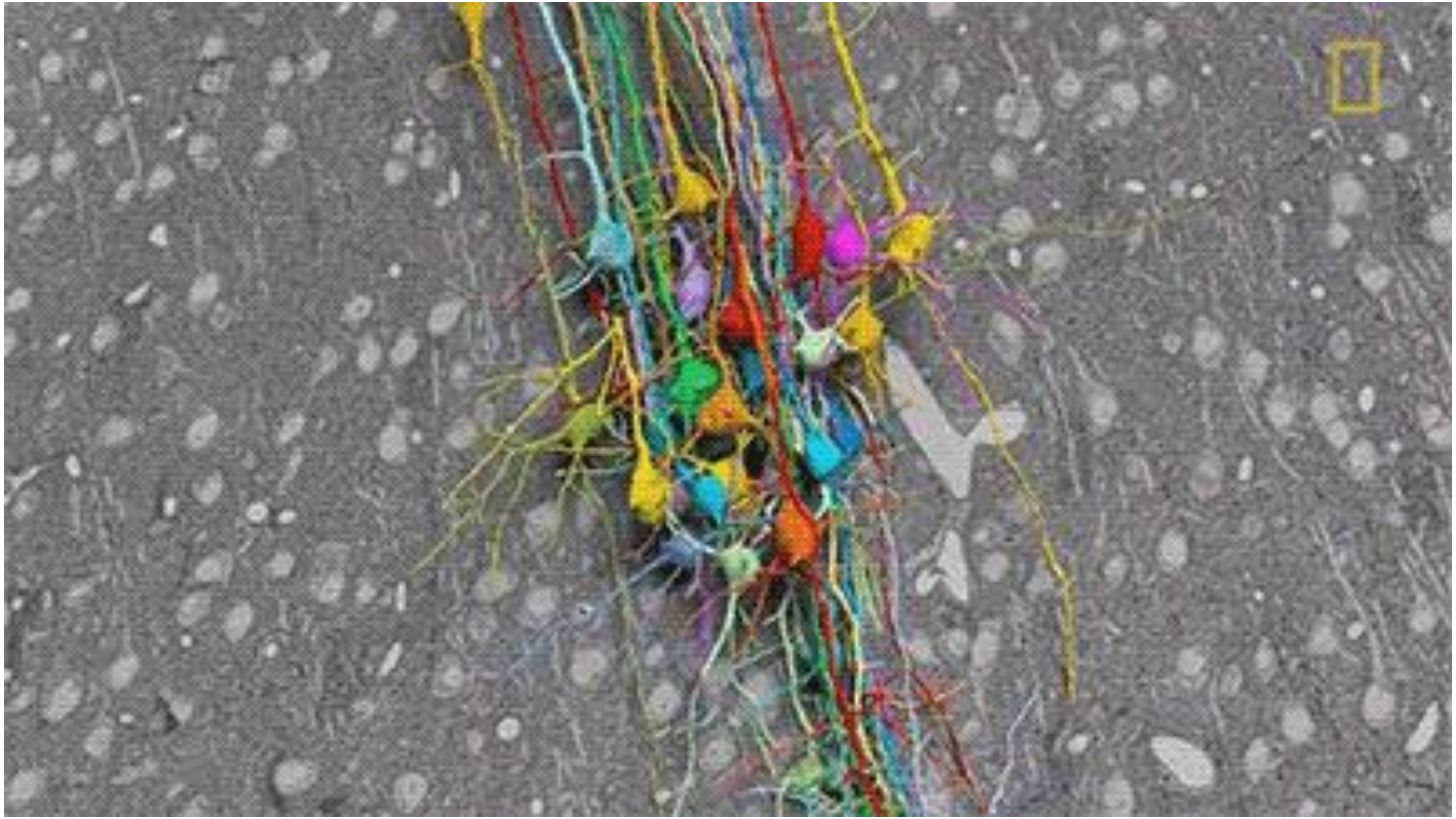
Neuron

I dendriti e il corpo cellulare sono ricoperti da centinaia di sinapsi, tramite le quali il neurone riceve informazioni da altri neuroni.

L'informazione raggiunge i bersagli sotto forma di segnali elettrici, tramite l'assone.

Ogni terminazione assonica forma una sinapsi con la quale trasferisce l'informazione a un'altra cellula.



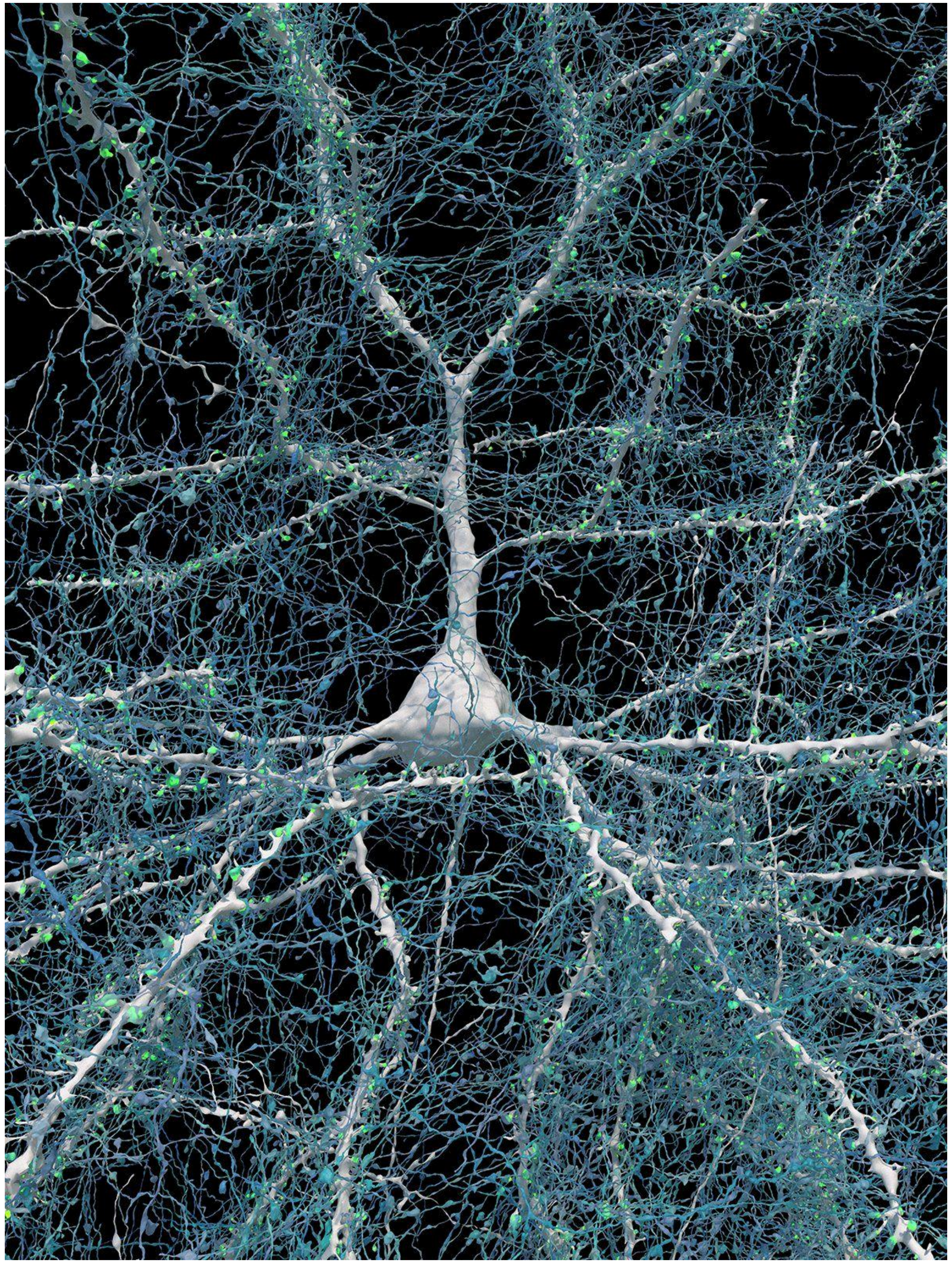


RESEARCH ARTICLE

NEUROSCIENCE

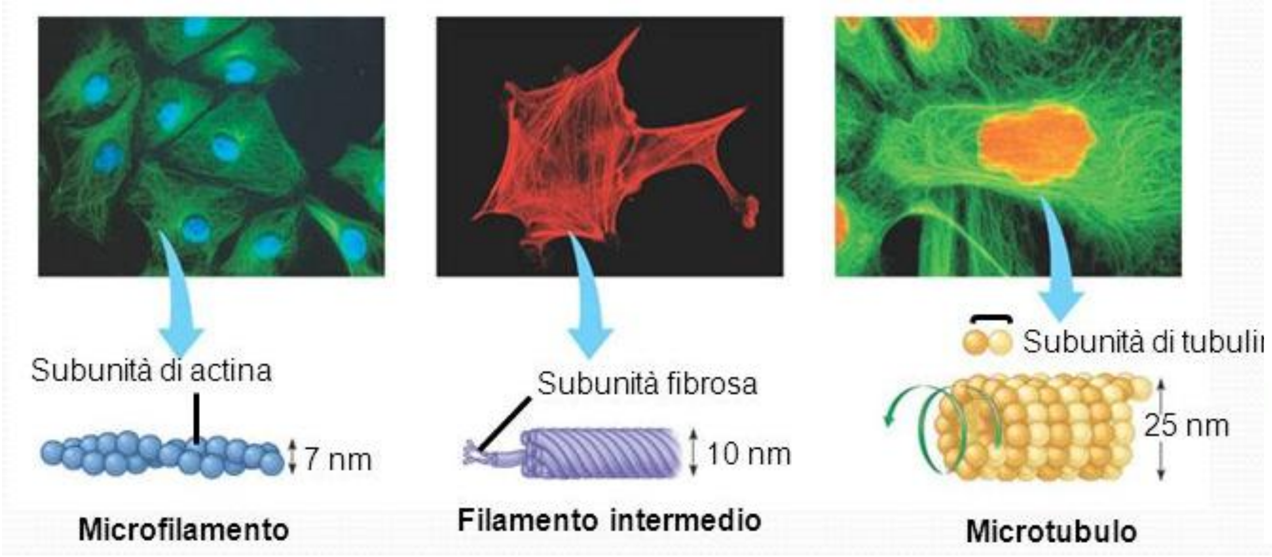
A petavoxel fragment of human cerebral cortex reconstructed at nanoscale resolution

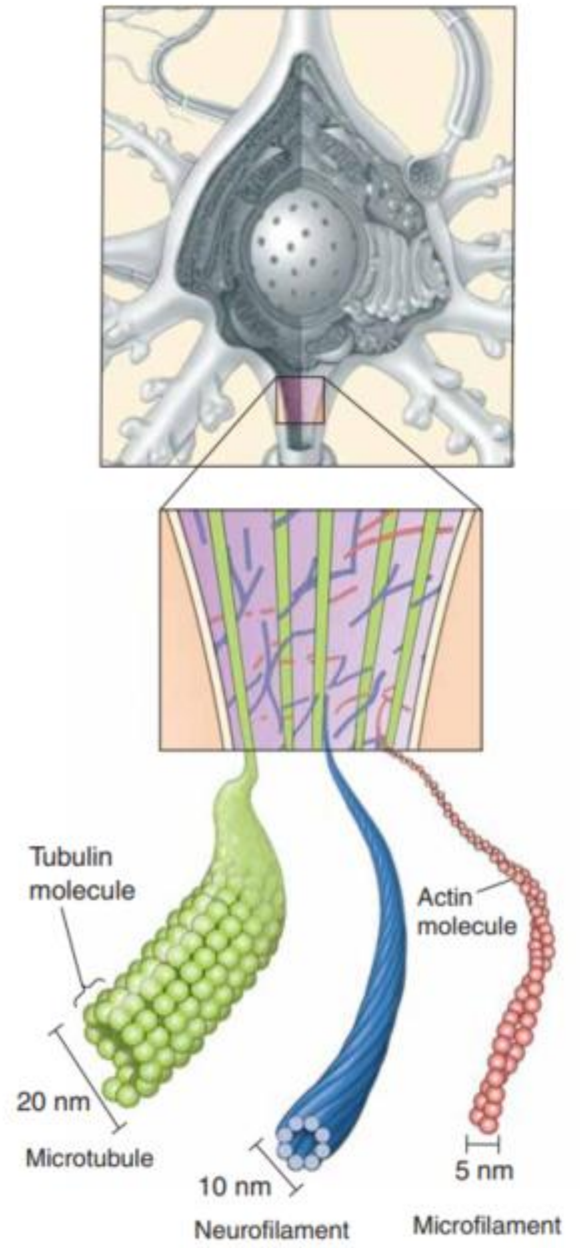
Alexander Shapson-Coe[1,2]†, Michał Januszewski[3]†, Daniel R. Berger[1]†, Art Pope[4], Yuelong Wu[1], Tim Blakely[5], Richard L. Schalek[1], Peter H. Li[4], Shuohong Wang[1], Jeremy Maitin-Shepard[4], Neha Karlupia[1], Sven Dorkenwald[4,6,7], Evelina Sjostedt[1], Laramie Leavitt[4], Dongil Lee[1,8], Jakob Troidl[9], Forrest Collman[10], Luke Bailey[1], Angerica Fitzmaurice[1,11], Rohin Kar[1,11], Benjamin Field[1,11], Hank Wu[1,11], Julian Wagner-Carena[1], David Aley[1], Joanna Lau[1], Zudi Lin[9], Donglai Wei[12], Hanspeter Pfister[9], Adi Peleg[1,13], Viren Jain[4]*, Jeff W. Lichtman[1]*



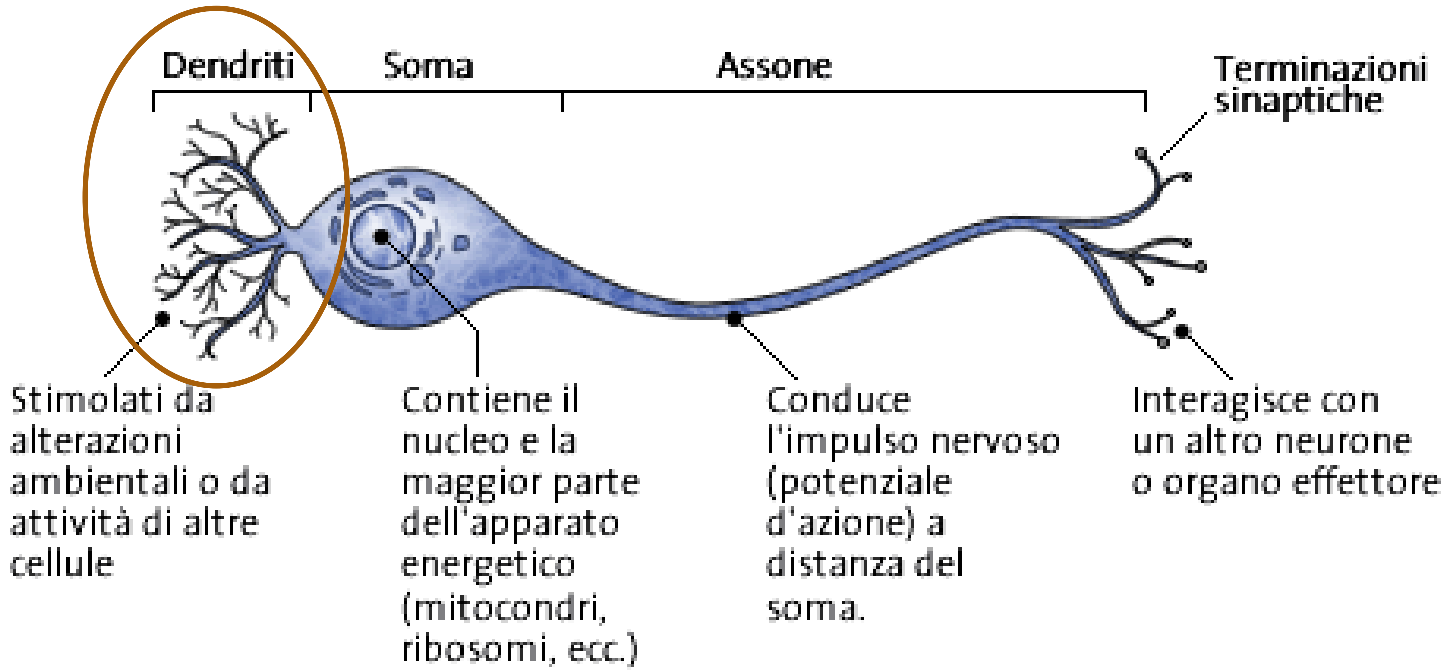
La struttura tridimensionale del neurone è mantenuta grazie al CITOSCHELETRO, una struttura proteica composta da:

microtubuli
microfilamenti
neurofilamenti



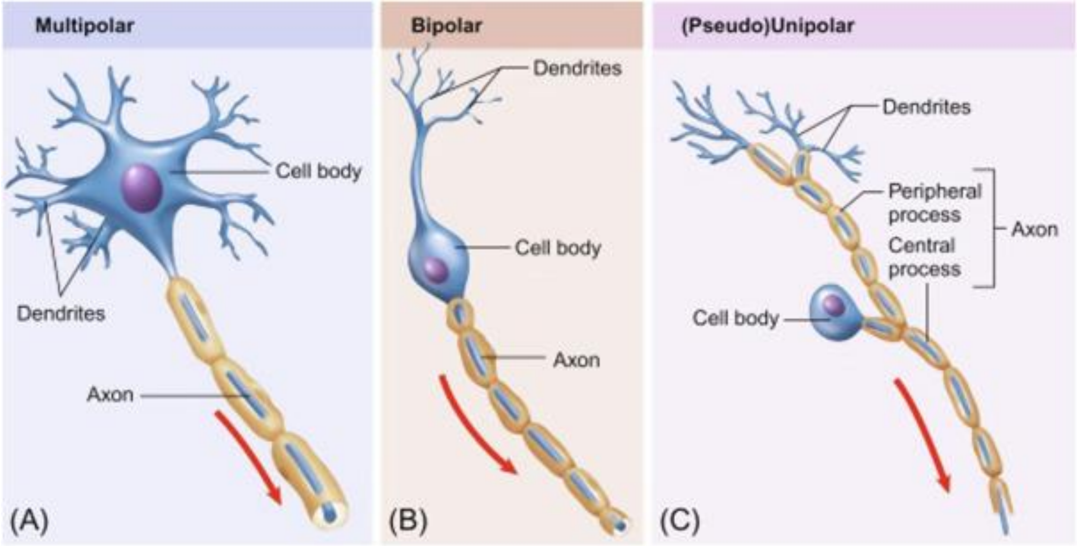


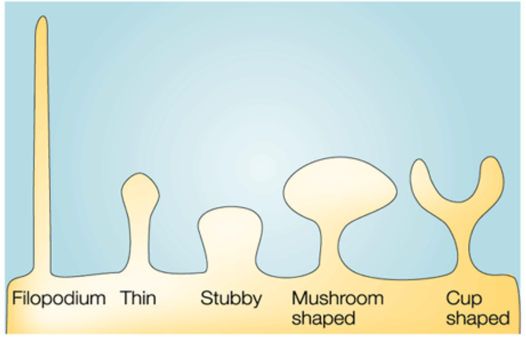
Dendrites



Dendriti
Soma
Assone
Terminazioni sinaptiche
Stimolati da alterazioni ambientali o da attività di altre cellule
Contiene il nucleo e la maggior parte dell'apparato energetico (mitocondri, ribosomi, ecc.)
Conduce l'impulso nervoso (potenziale d'azione) a distanza del soma.
Interagisce con un altro neurone o organo effettore

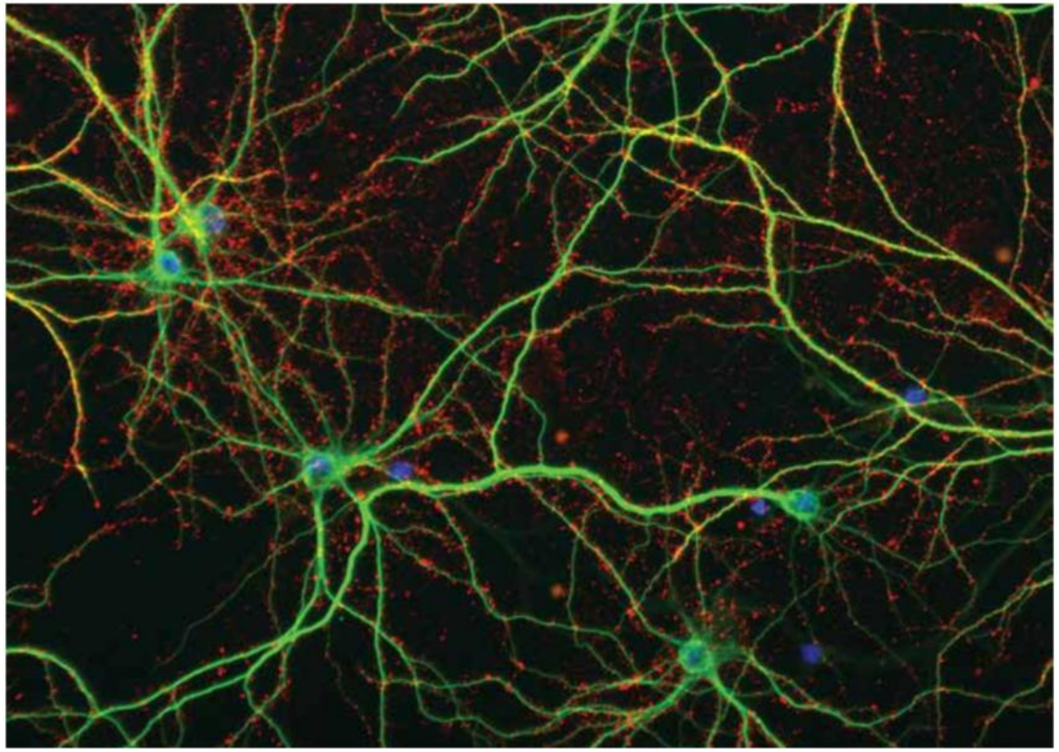
Various type of dendrites (and neurons)

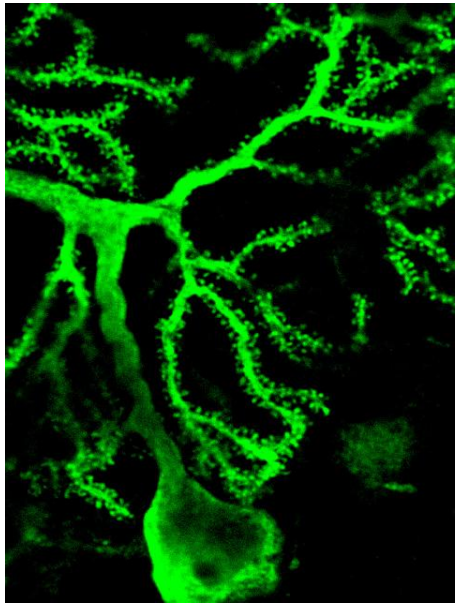


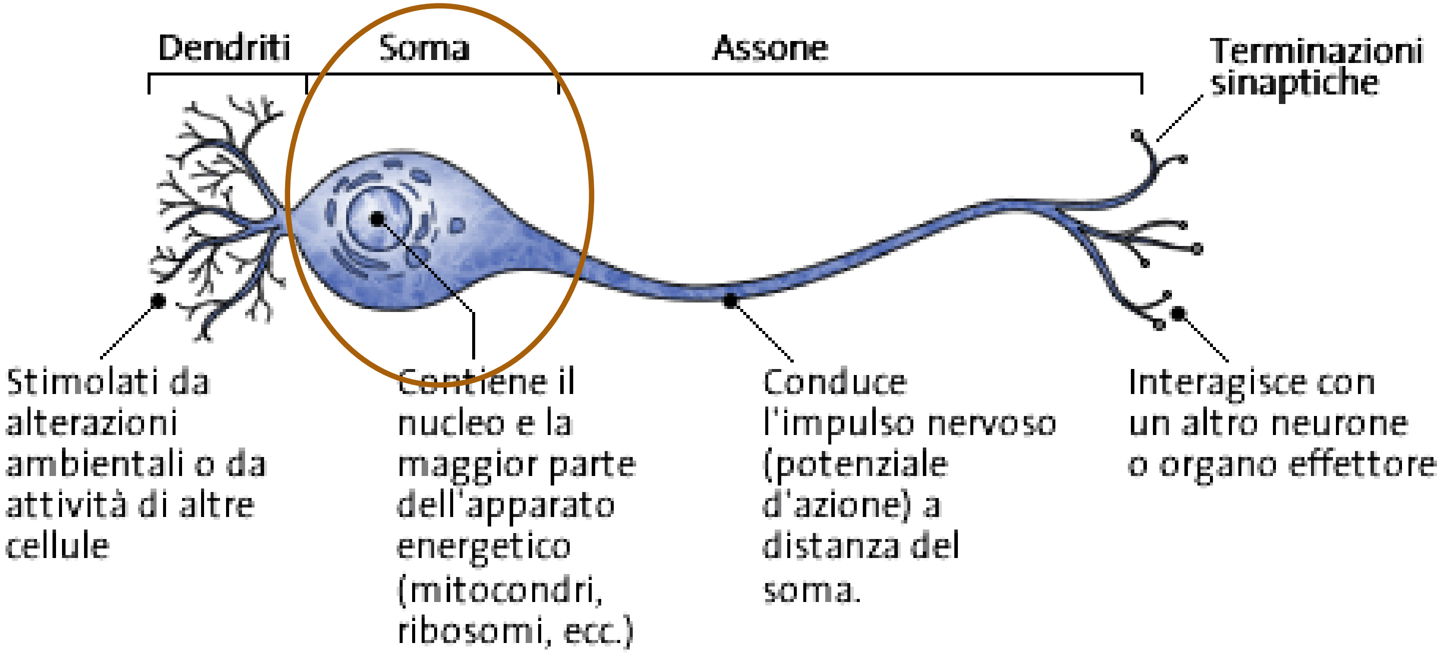


Nature Reviews | Neuroscience

Dendritic Spines



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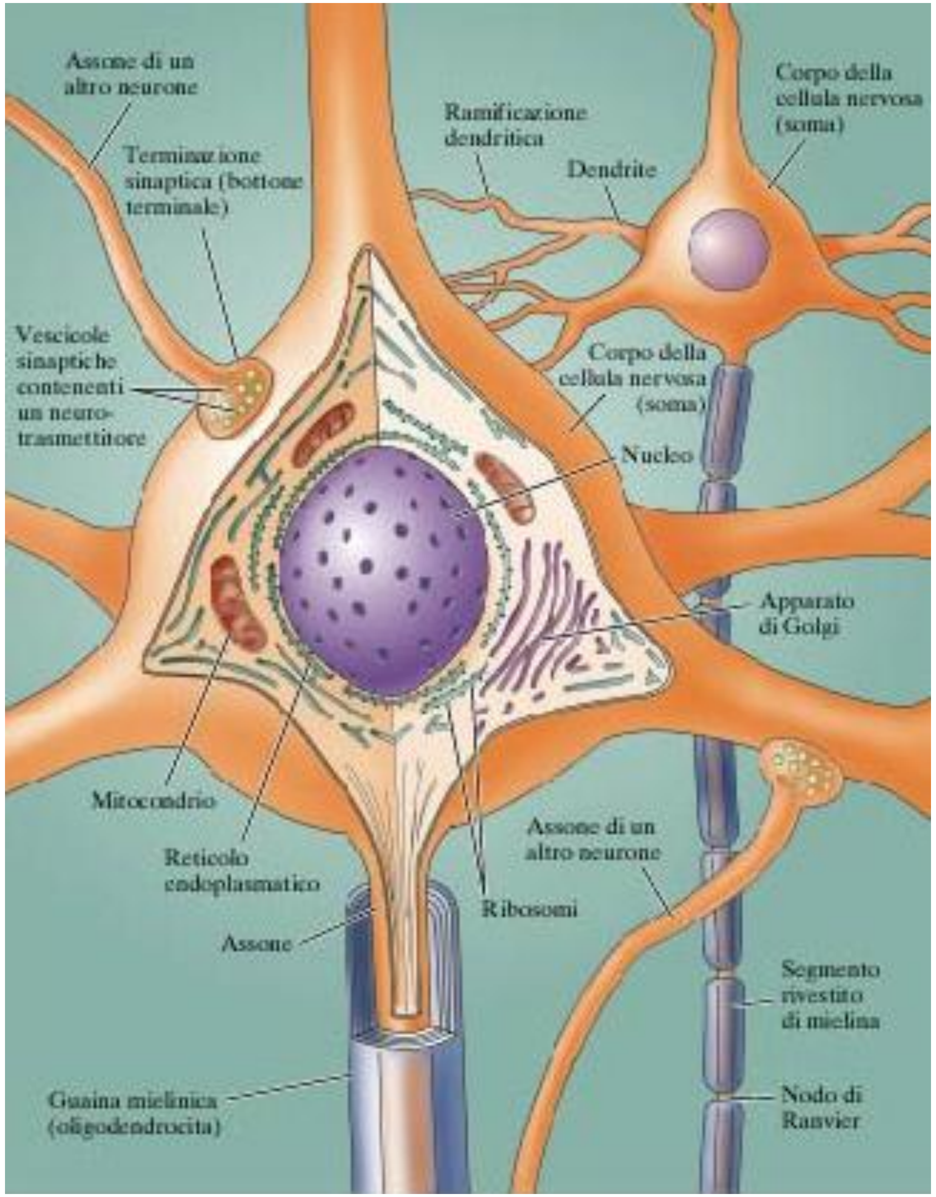
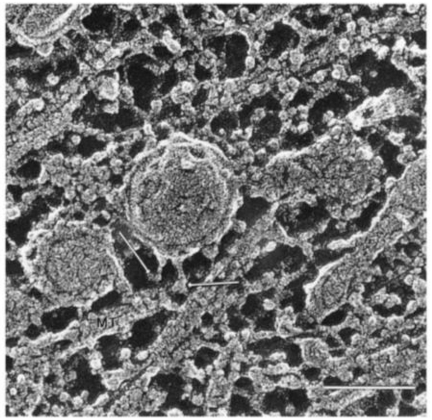
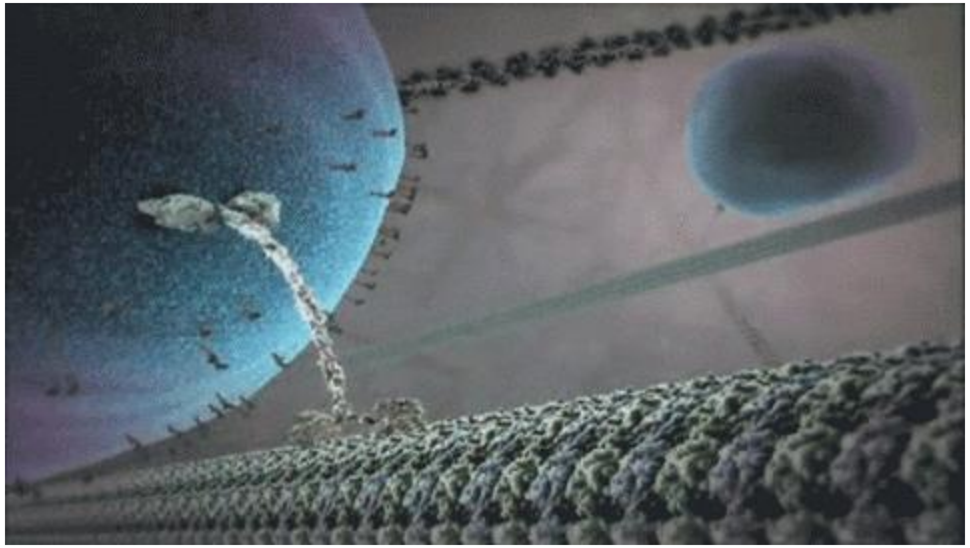
Most biosynthetic products are made in the soma

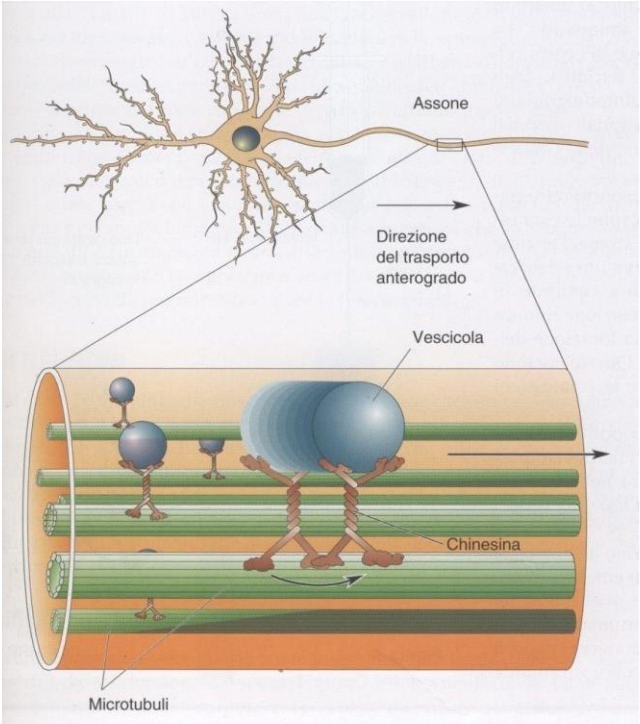


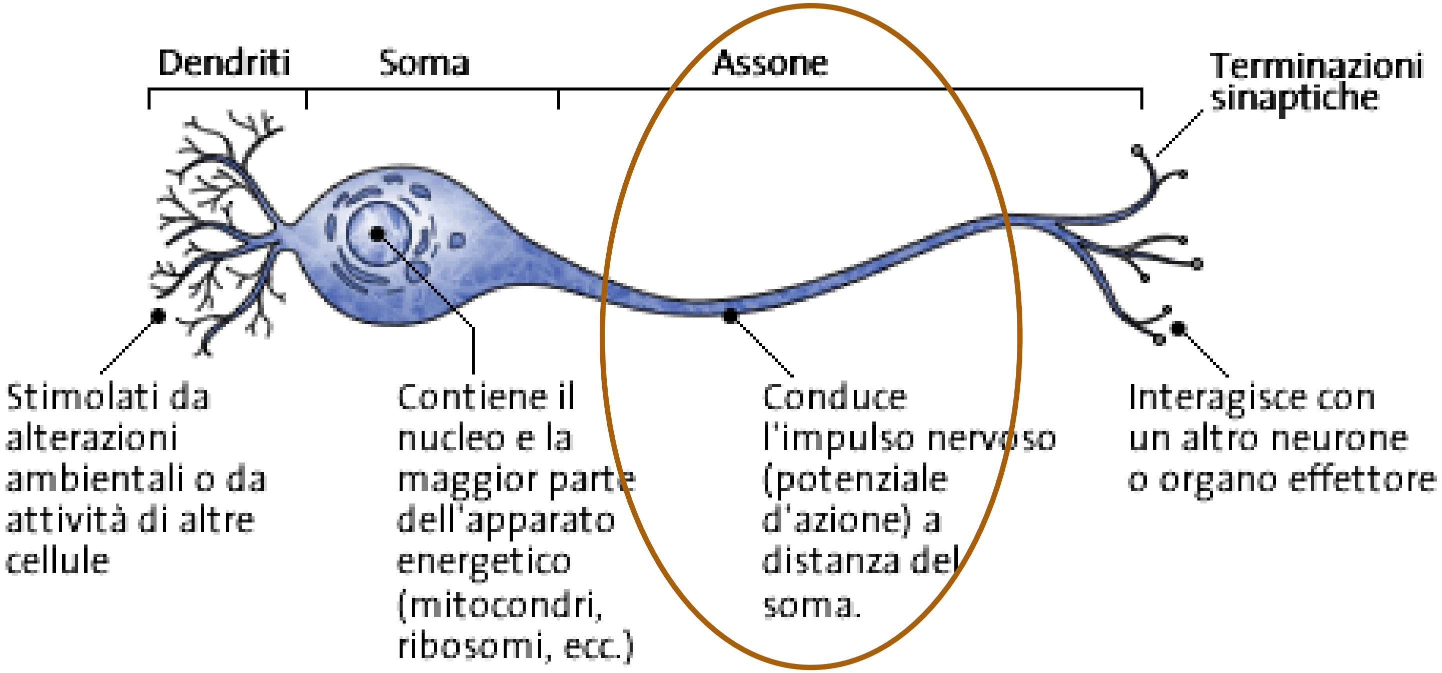
One Option - Rapid axonal transport

- fast, microtubule-based movement of vesicles/organelles along axons—anterograde via kinesin, retrograde via dynein (≈50–400 mm/day).





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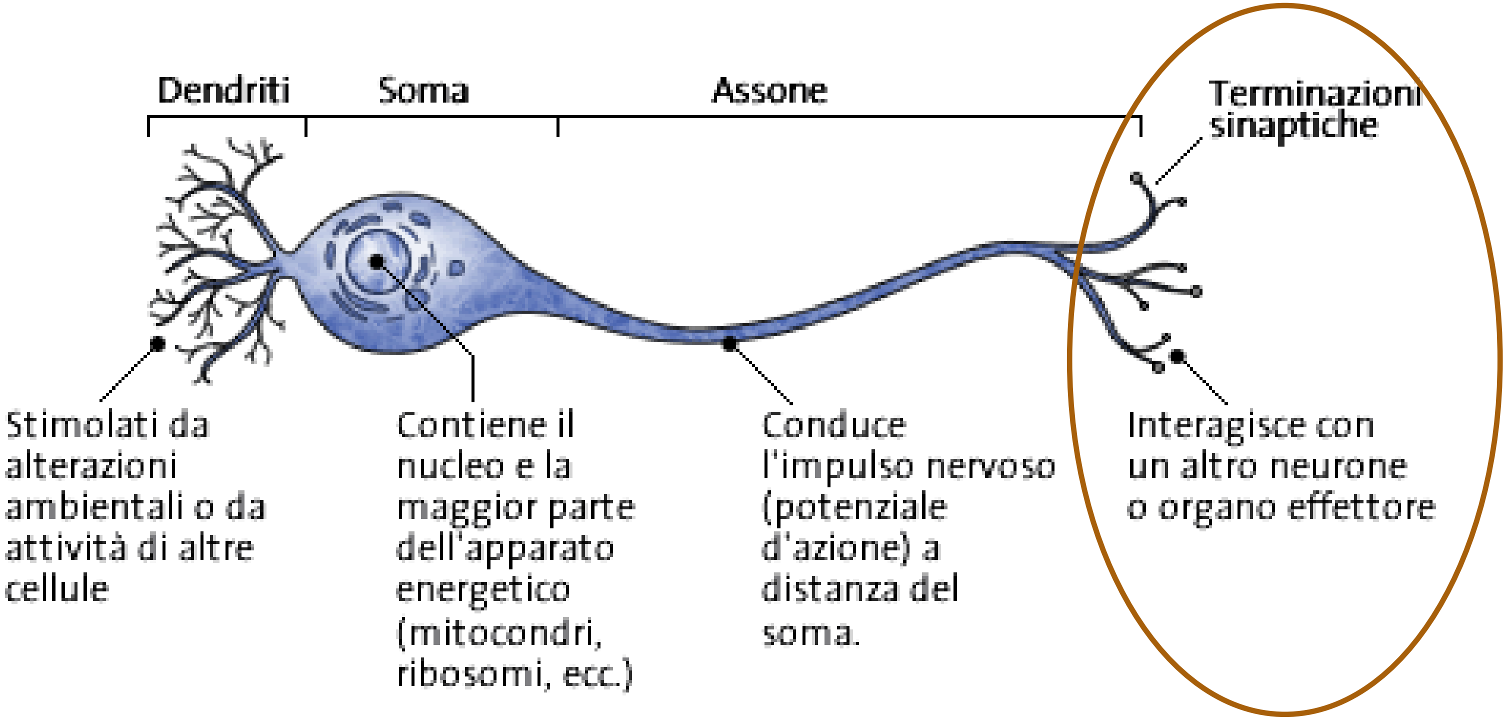
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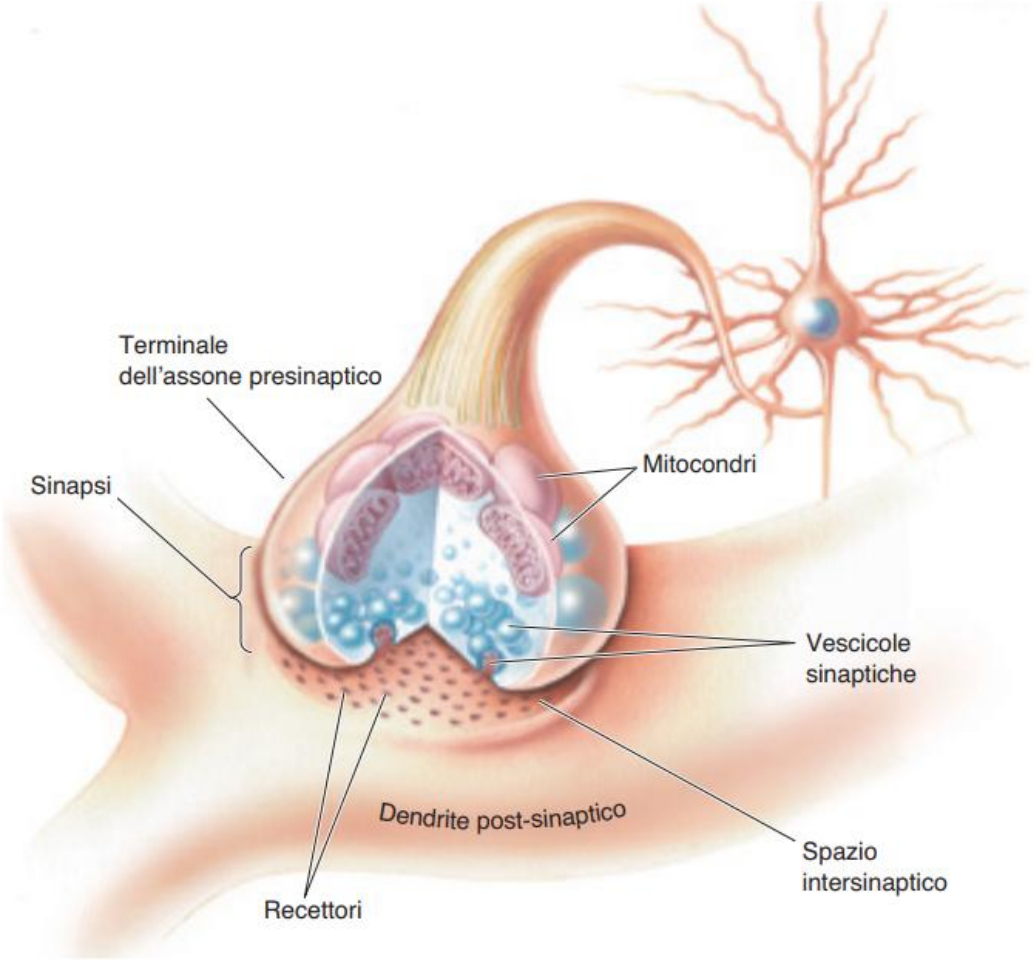
Chemical Synapse

Presynaptic terminal (bouton): synaptic vesicles, active zone, voltage-gated Ca^{2+} channels

Synaptic cleft: extracellular matrix/basal lamina, cell-adhesion molecules (e.g., neurexin–neuroligin), enzymes (e.g., AChE at NMJ).

Postsynaptic side: postsynaptic density (PSD) with neurotransmitter receptors (ionotropic & metabotropic) and scaffold proteins.



Classification of neurons

The human nervous system contains **over 100 billion neurons** of **at least 10,000 different types:**

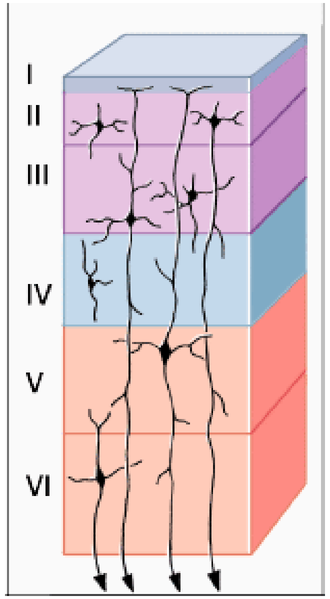
- **According to the number of processes** that originate from the soma

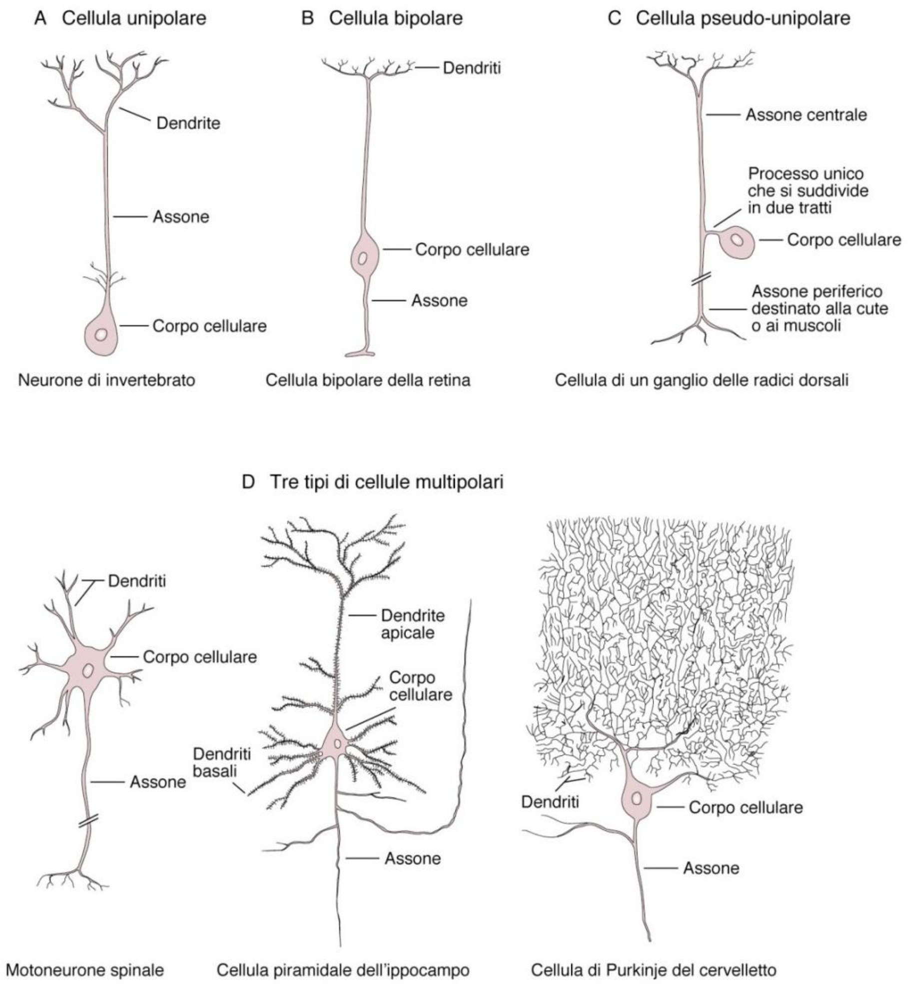
- **According to function:**

Sensory neurons
Motor neurons
Interneurons

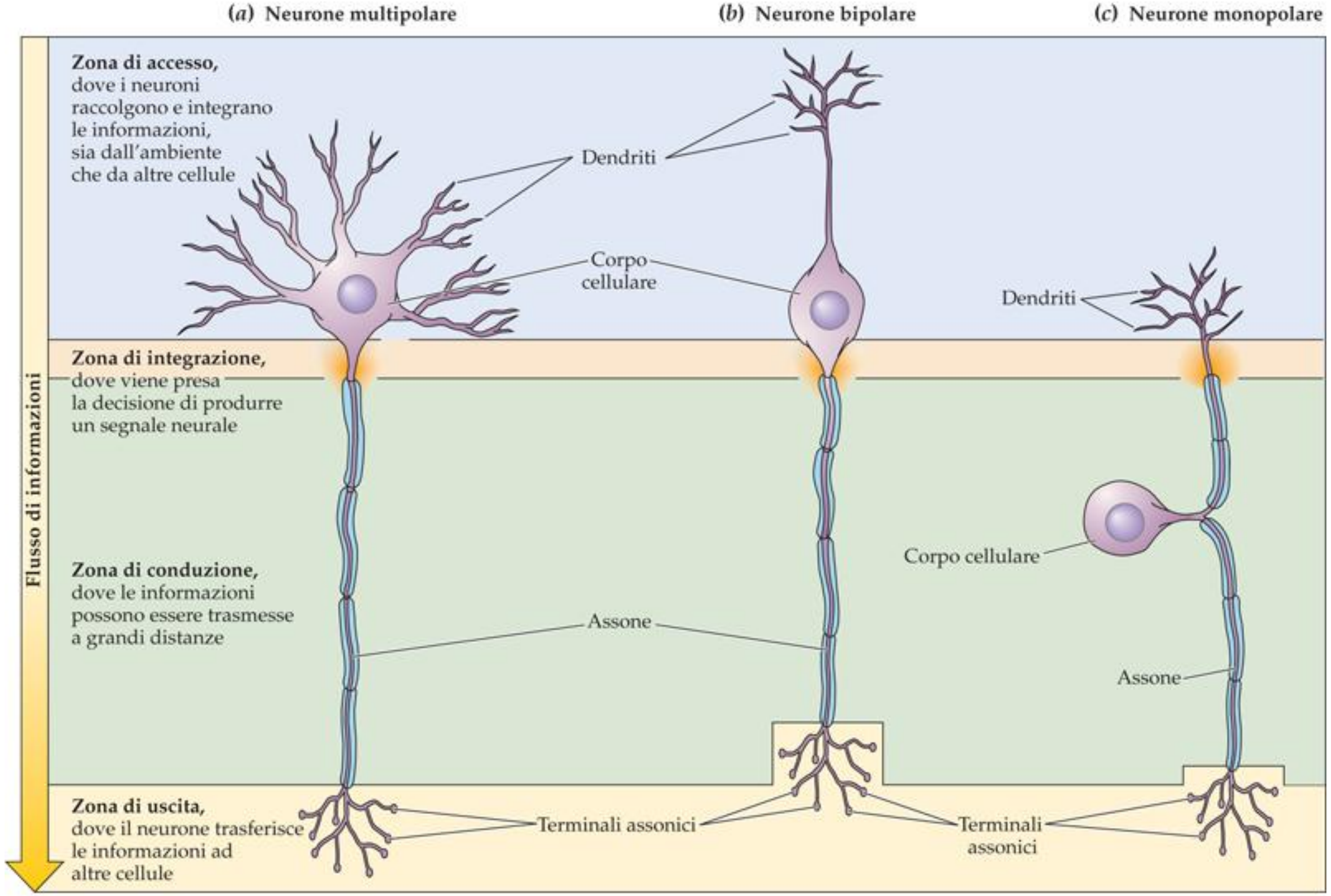
- **In the CNS (central nervous system):**

Local interneurons
Projection neurons

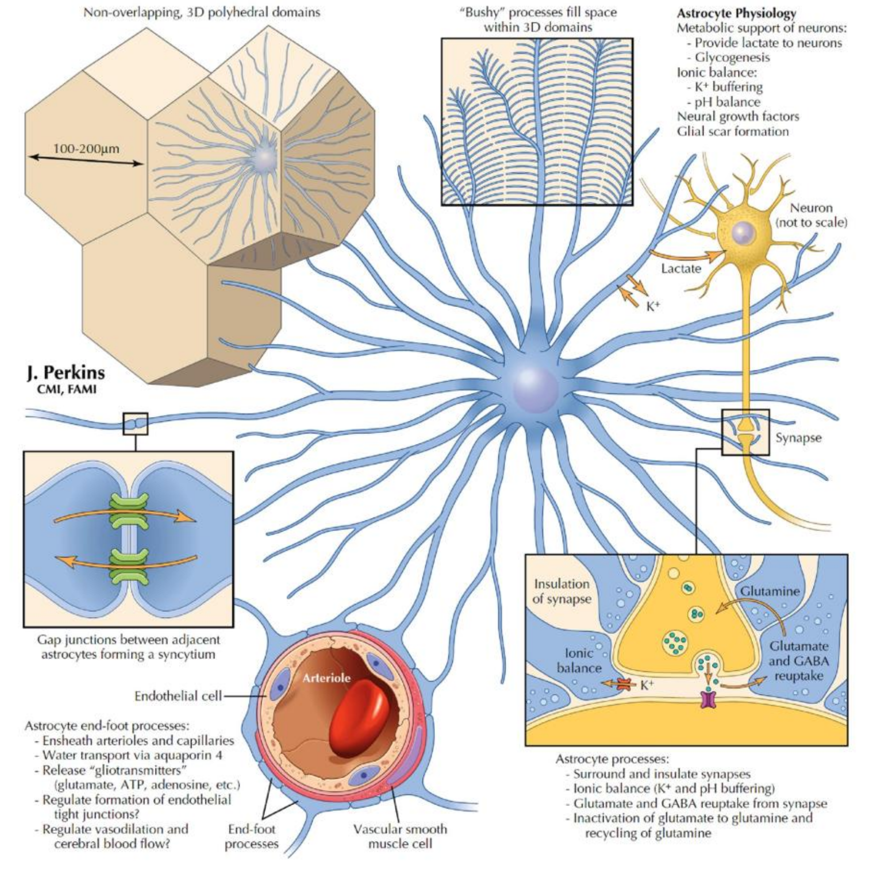




Functional components of the neuron



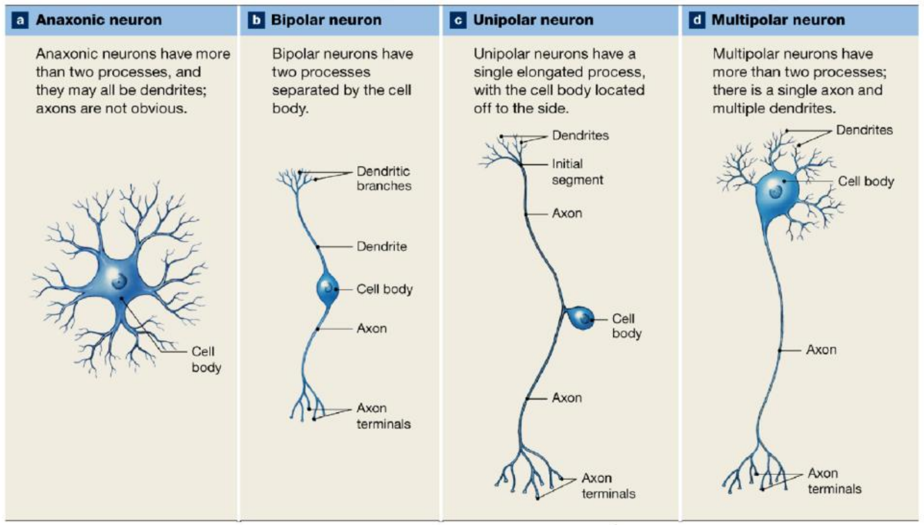
Neurons and Their Properties

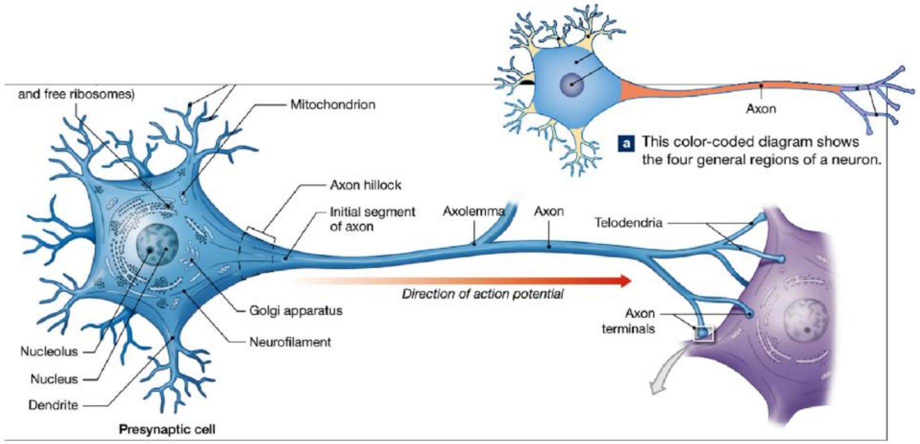


ASTROCYTE BIOLOGY

Astrocytes are the most abundant glial cells in the CNS. They arise from neuroectoderm and are intimately associated with neural processes, synapses, vasculature, and the pial-glial membrane investing the CNS. Astrocytes in gray matter are called *protoplasmic astrocytes*, and in white matter they are called *fibrous astrocytes*. The somas vary in diameter from a few µm to 10 or more µm. Astrocytes are arrayed in non-overlapping 3D polyhedral domains of 100-200 µm across (up to 400 µm in hominids). Structurally, astrocytic processes interdigitate, forming a syncytium to protect synapses (as close as 1µm to these structures). Astrocytic endfeet associate with vascular endothelial cells and associated smooth muscle cells. Astrocytic processes invest the entire pial membrane from the inside.

Physiologically, astrocytic processes affect ion balance (sequester K^+), transport water via aquaporin 4 channels, uptake and recycle glutamate and GABA, provide metabolic support to neurons, and can become reactive after CNS injury and lay down glial scar tissue. Astrocytes also can release growth factors and bioactive molecules (termed *gliotransmitters*) such as glutamate, ATP, and adenosine. In development, specialized astrocytes, called *radial glia*, provide a scaffold for orderly neural migrations in the CNS.



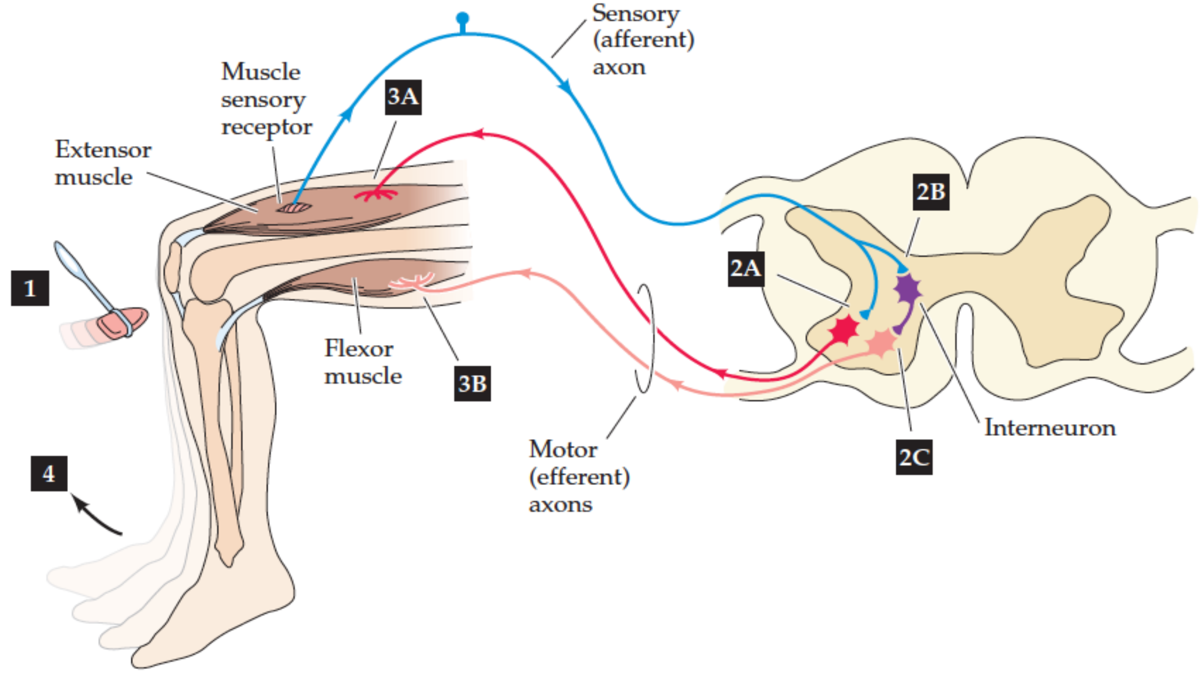


Neurons are nerve cells specialized for intercellular communication

Source : Netters Atlas of Neuroscience, 3e by Felten, David L. Maida, Mary Summo Netter, Frank Henry OBanion, M. Kerry

Example of a neural circuit

Myotatic (stretch) reflex



1 Hammer tap stretches tendon, which, in turn, stretches sensory receptors in leg extensor muscle

2 (A) Sensory neuron synapses with and excites motor neuron in the spinal cord

(B) Sensory neuron also excites spinal interneuron

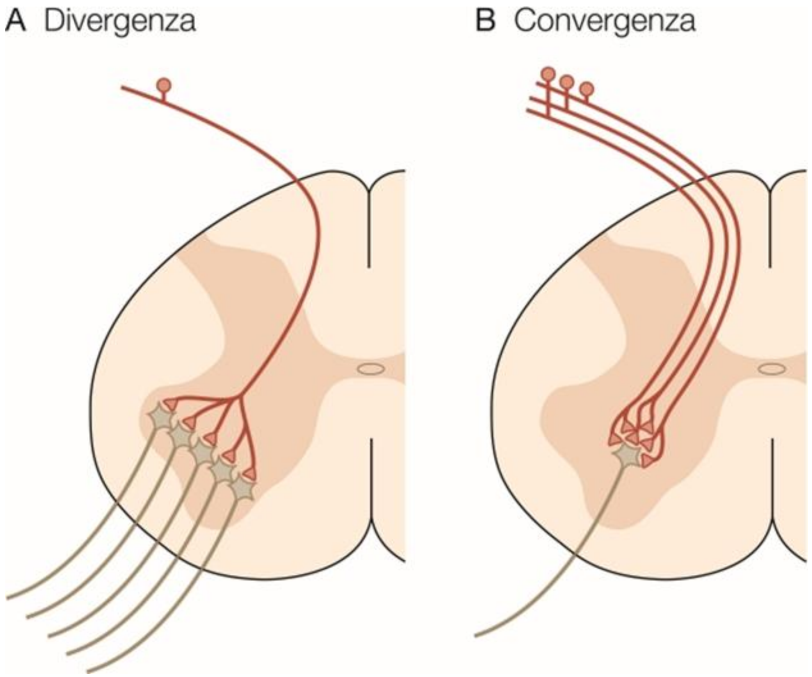
(C) Interneuron synapse inhibits motor neuron to flexor muscles

3 (A) Motor neuron conducts action potential to synapses on extensor muscle fibers, causing contraction

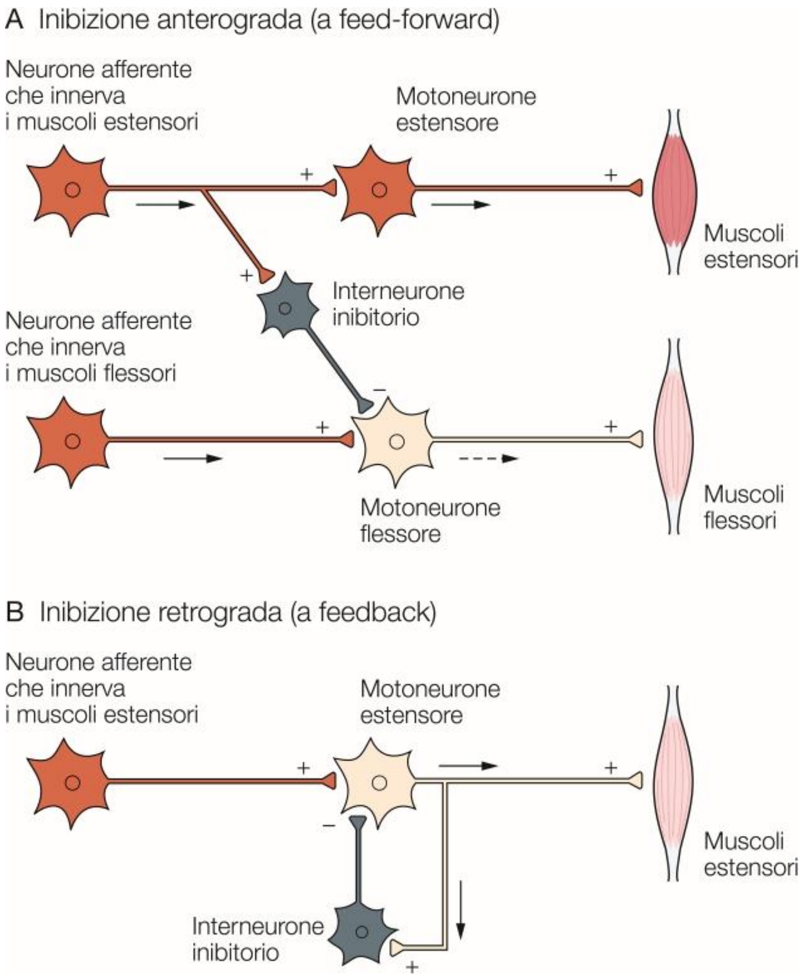
(B) Flexor muscle relaxes because the activity of its motor neurons has been inhibited

4 Leg extends

Divergence vs Convergence



Inibition

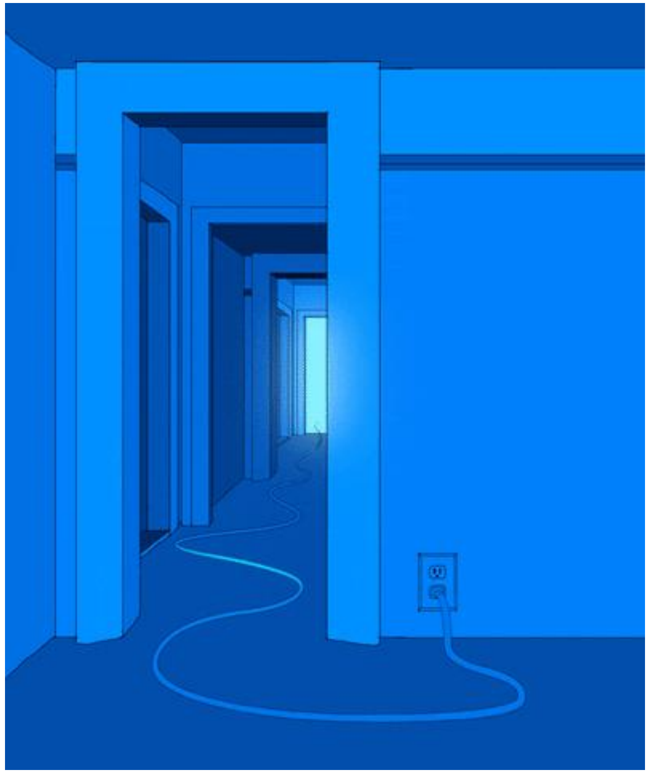


Neuron is bad

Problems related to the use of electrical signals:

- Axons are not good conductors of electricity.
- Neurons conduct electricity passively.

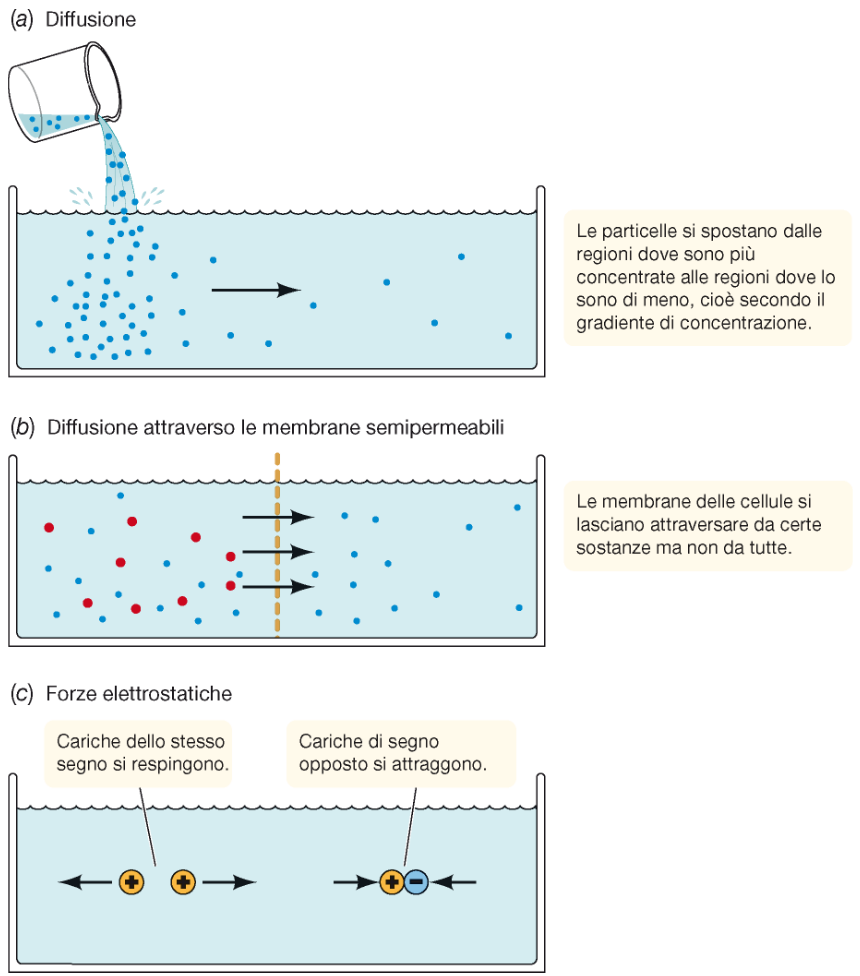
"Amplifier system" that allows electrical signals to be conducted over long distances **(Action Potential).**



Ions motion

Diffusion (gradient)

Electrostatic force



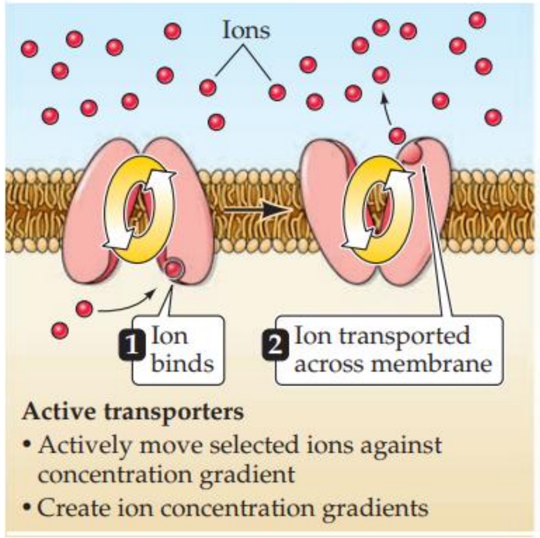
Some Definitions

- **Electrical current (I)** = flow of charged particles (ions) (**amperes**)
- Force experienced by a charged particle (**electric potential**) = **voltage** (V)
- **Electrical conductance** (g) = a measure of how easily an ion can cross the cell membrane
- The inverse of conductance is **resistance** (R) = a measure of how difficult it is for an ion to cross the cell membrane

How ion movements produce electrical signals

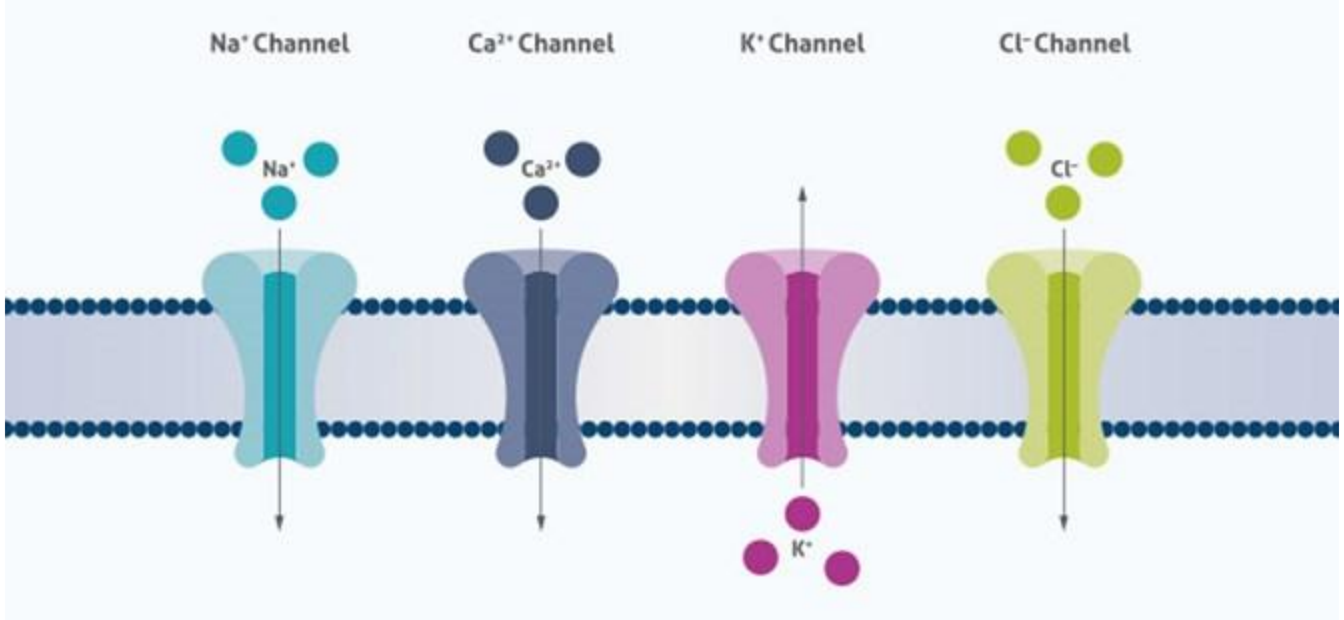
Electrical potentials are generated across neuronal membranes because:

1) There are **differences in the concentration** of specific ions on the two sides of the membrane → These concentration gradients are also established by **active transporters**, that is, proteins that actively move ions into or out of cells in the opposite direction to their gradients.



How ion movements produce electrical signals

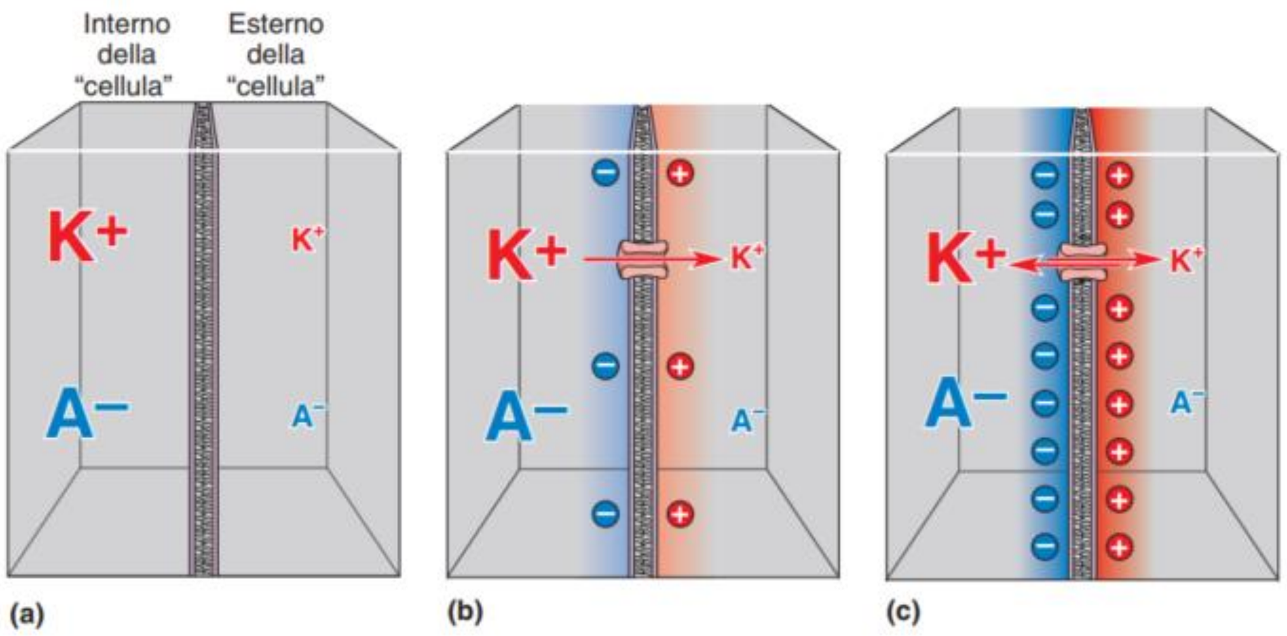
2) Permeability is largely due to **ion channels**, which are proteins that allow only certain types of ions to cross the membrane, moving in the same direction as their gradients.



Channels and transporters work against each other

Generation of membrane potentials

When there is a perfect balance between the two opposing forces, an **electrochemical equilibrium** is reached.



Forces that generate membrane potentials

The electrical potential difference that arises across the membrane under conditions of electrochemical equilibrium (the *equilibrium potential*) can be predicted using <u>the Nernst equation.</u>

It calculates the **voltage value** necessary to exactly balance the concentration difference of an ion across the membrane.

$$E_{cell} = E^0 - \left(\frac{RT}{nF}\right) \ln Q$$

Ecell: the actual potential of the electrochemical cell under given conditions.
E0: the standard cell potential, measured when all reactants/products are at standard conditions.
R: universal gas constant.
T: absolute temperature (Kelvin).
n: number of moles of electrons transferred in the redox reaction.
F: Faraday's constant.
Q: the reaction quotient, ratio of products to reactants at any moment, raised to the power of their stoichiometric coefficients.

Equilibrium potential for potassium:

$$E_{K+} = -90\ \text{mV}$$

Resting membrane potential:

$$V_m \approx -70\ mV$$

At rest, the membrane has many more open potassium channels than sodium channels (≈30:1).

As a result, the **resting potential** is much closer to the potassium equilibrium potential (−90 mV) than to the **sodium equilibrium potential** (+55 mV).

Electrochemical equilibrium in an environment with multiple permeant ions.

When more than one type of ion can cross the membrane, the membrane potential is no longer determined by a single Nernst equilibrium.

Goldman–Hodgkin–Katz (GHK) equation

$$V_m = \frac{RT}{F} \ln\left(\frac{p_K[K^+]_o + p_{Na}[Na^+]_o + p_{Cl}[Cl^-]_i}{p_K[K^+]_i + p_{Na}[Na^+]_i + p_{Cl}[Cl^-]_o} \right)$$

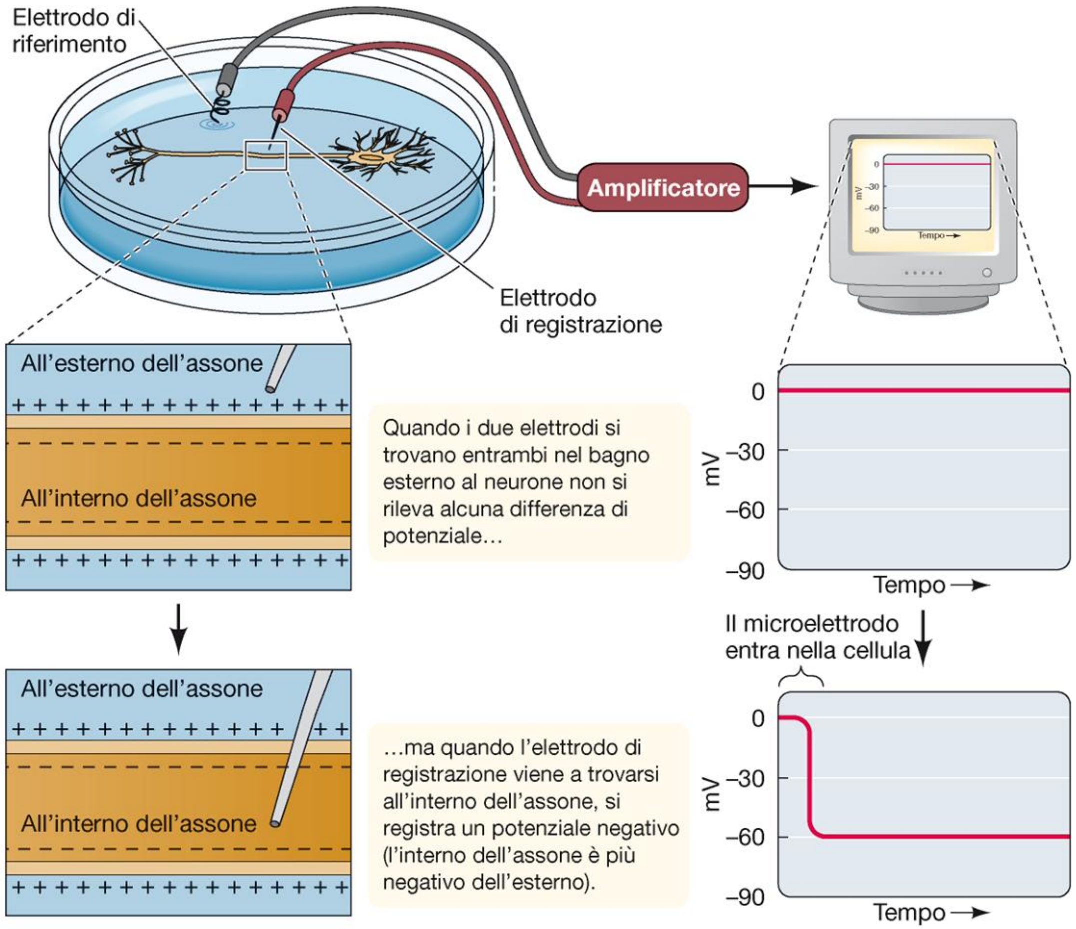
P = permeability of the ion
I = intracellular concentration
O = extracellular concentration

But what happens if we disturb the equilibrium?

The transmembrane electrical potentials of nerve cells

<u>Observation of the electrical difference using a microelectrode</u>

Resting membrane potential: the cell has a system capable of generating a constant potential difference when at rest (between −40 and −90 mV).

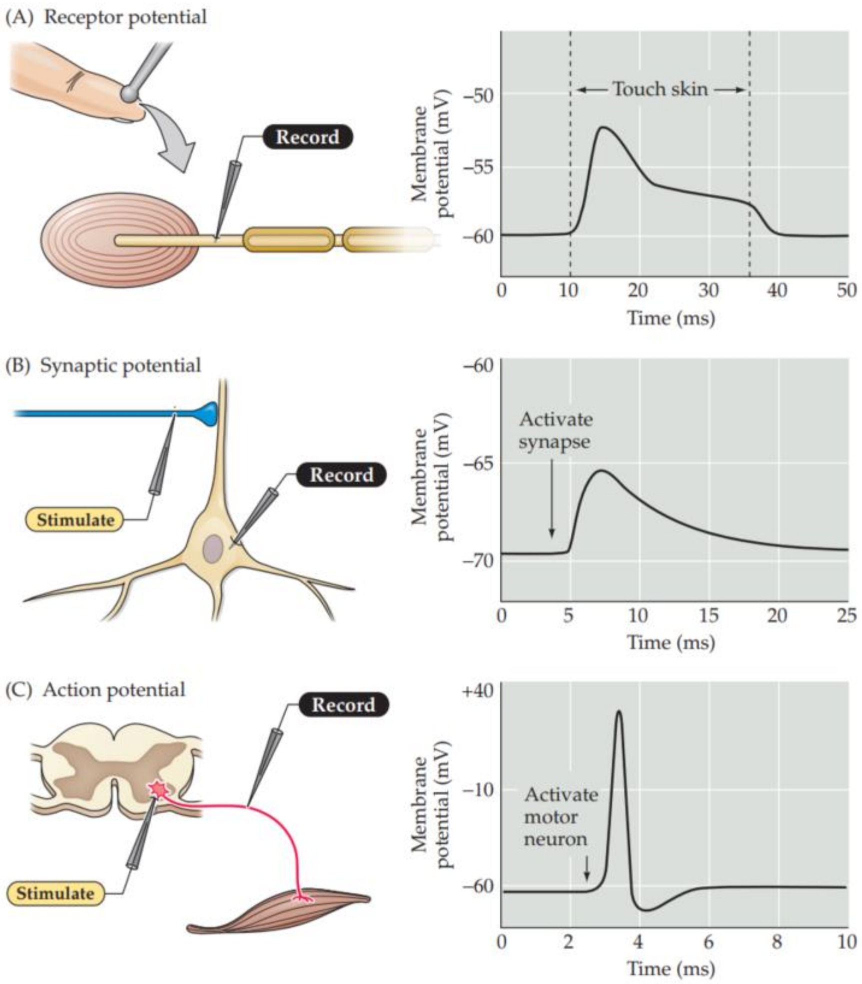


The transmembrane electrical potentials of nerve cells

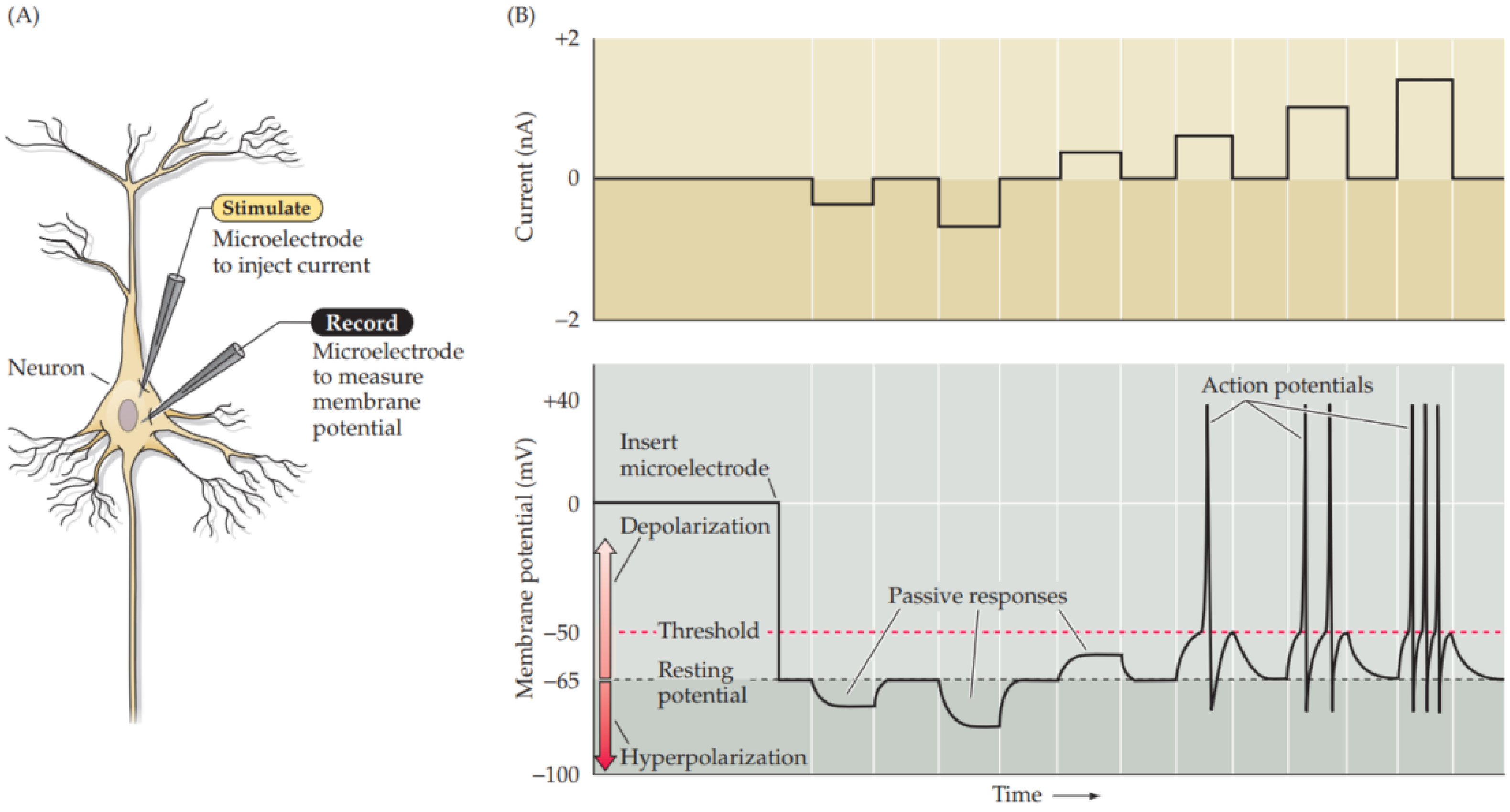
Receptor potentials: caused by the activation of sensory neurons induced by external stimuli

Synaptic potentials: associated with communication between neurons at the synaptic level (information transfer)

Action potential: potential actively generated by the neuron



Passage of an electrical current through the membrane of a neuron (in the laboratory via a second microelectrode)



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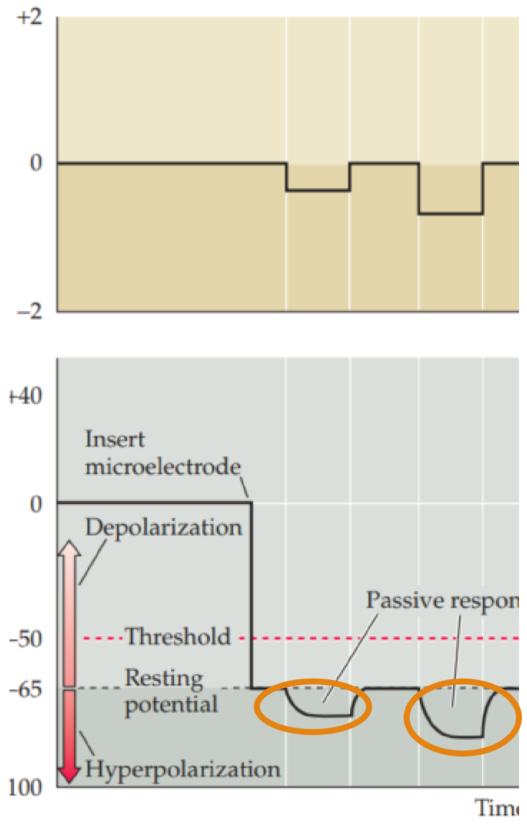


Hyperpolarization

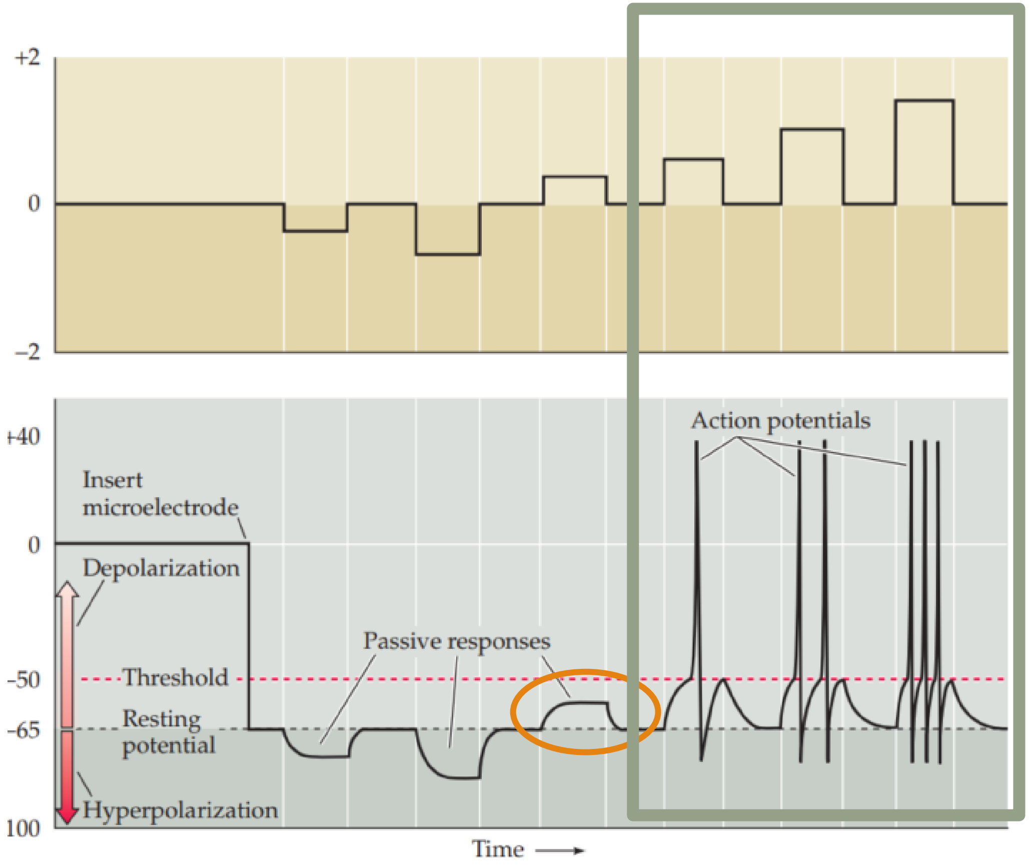
- The current produced makes the membrane potential more negative.
- The membrane potential changes in proportion to the amount of current injected.
- These are **passive electrical responses**.

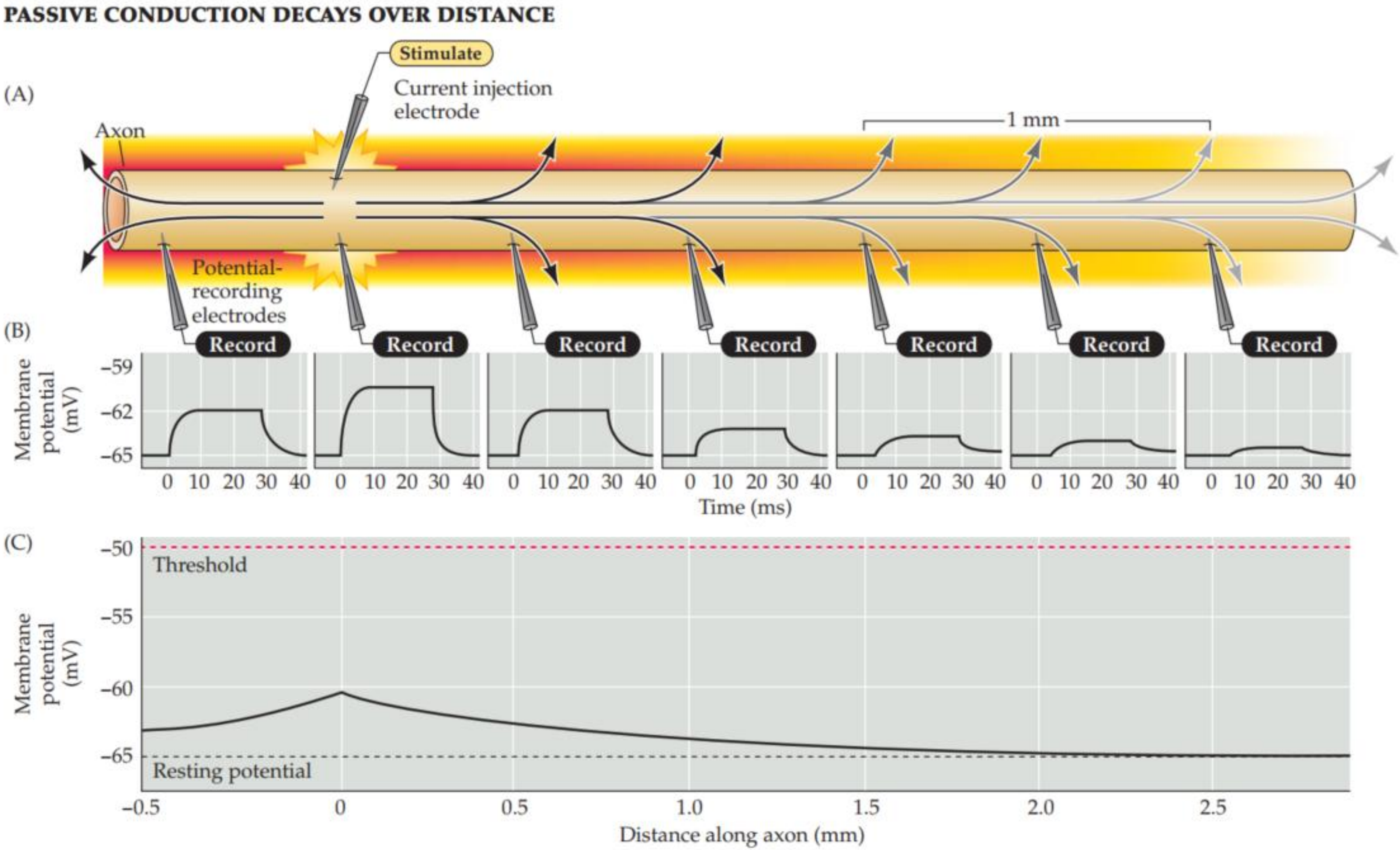
Passage of an electrical current through the membrane of a neuron (in the laboratory via a second microelectrode)

Depolarization

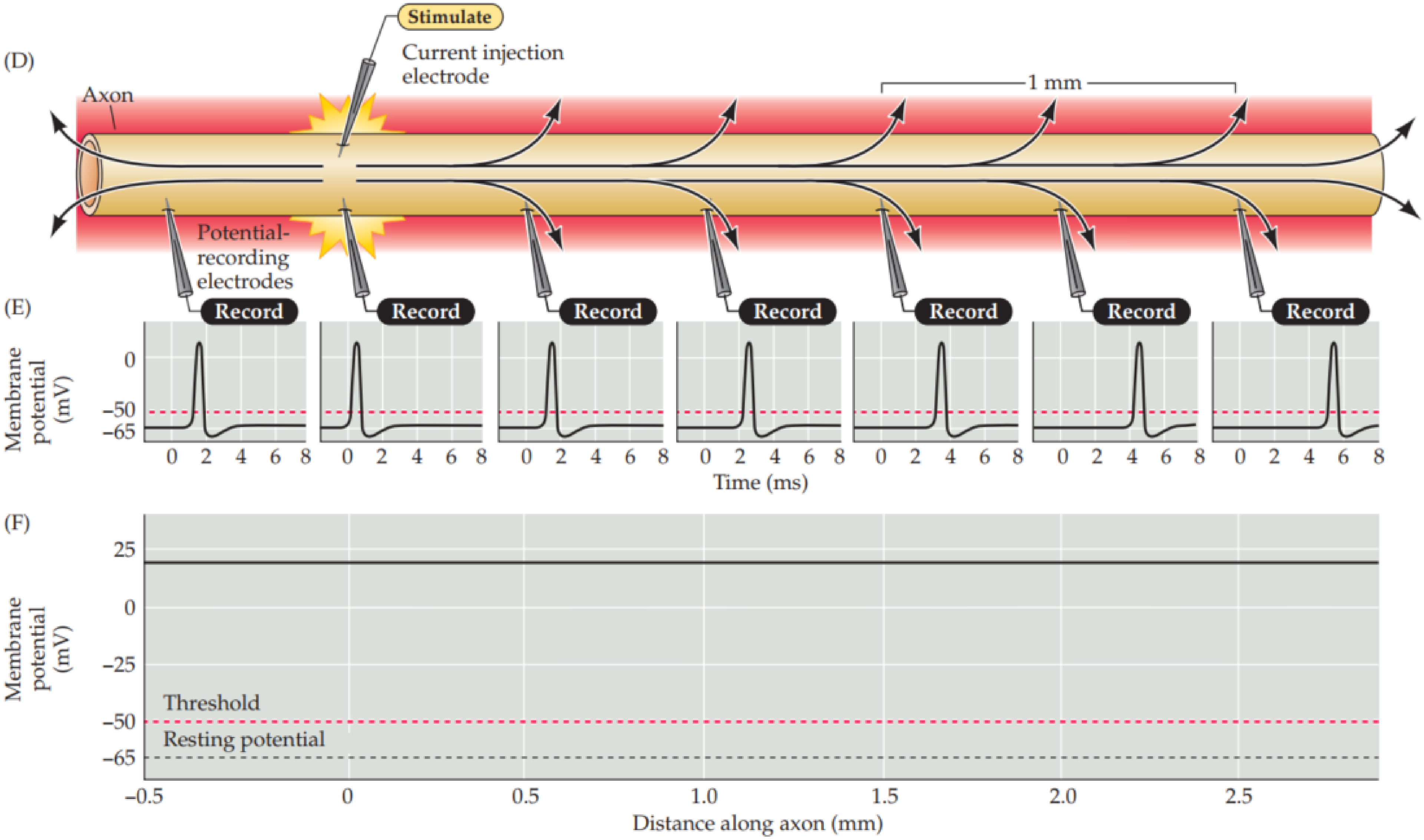
- Current of opposite polarity is injected.
- The membrane potential becomes more positive than the resting potential (and surpasses the threshold potential).

These are **active electrical responses (action potential)**.




PASSIVE CONDUCTION DECAYS OVER DISTANCE
(A)
Stimulate
Current injection electrode
Axon
1 mm
Potential-recording electrodes
(B)
Record
Record
Record
Record
Record
Record
Record
Membrane potential (mV)
−59
−62
−65
0 10 20 30 40
Time (ms)
(C)
Membrane potential (mV)
−50
−55
−60
−65
Threshold
Resting potential
−0.5
0
0.5
1.0
1.5
2.0
2.5
Distance along axon (mm)

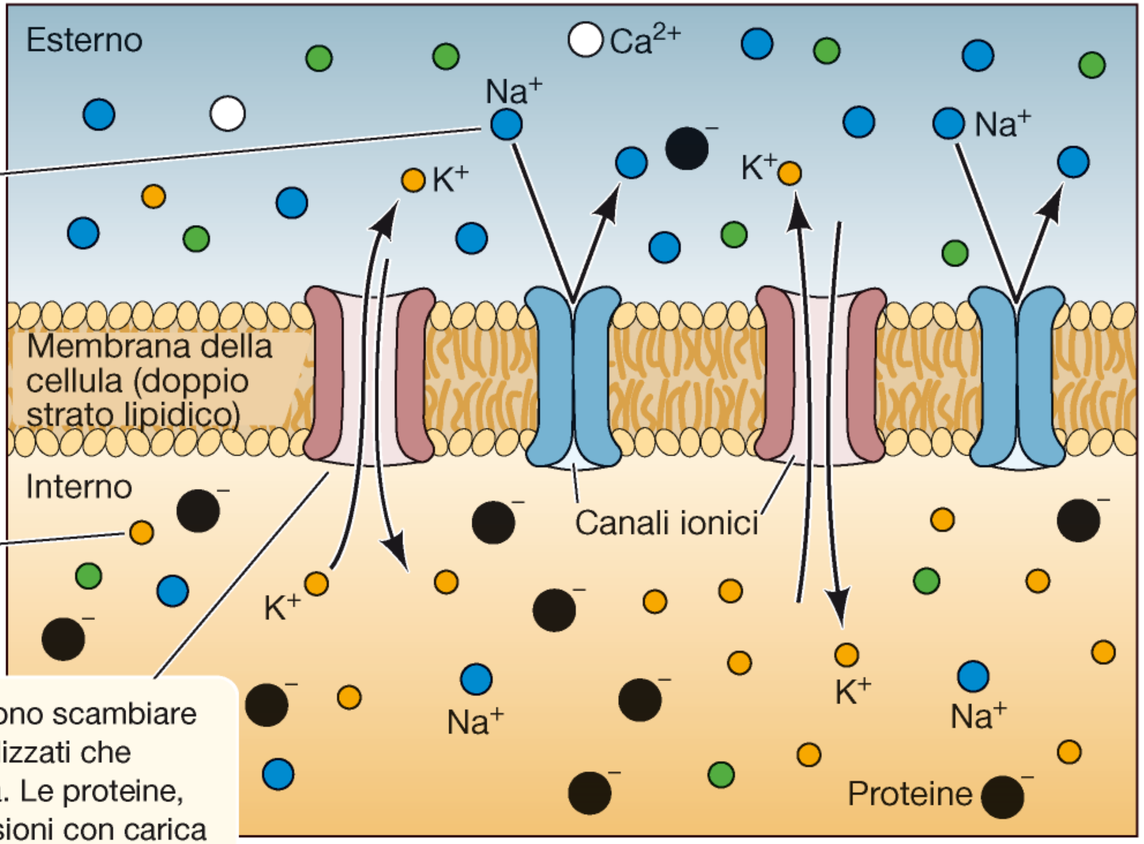
ACTIVE CONDUCTION IS CONSTANT OVER DISTANCE



Farinella, M. & Ros, H. (2014). Neurocomic. Rizzoli Lizard

	Na^+	K^+	Cl^-	Ca^{2+}	Proteine
All’esterno della cellula	molti	pochi	molti	molti	pochi
All’interno della cellula	pochi	molti	pochi	pochi	molti

La maggior parte degli ioni sodio (Na^+), cloruro (Cl^-) e calcio (Ca^{2+}) si trova nello spazio extracellulare.



La maggior parte degli ioni potassio (K^+) si trova all’interno del neurone.

L’interno e l’esterno possono scambiare ioni grazie a canali specializzati che attraversano la membrana. Le proteine, molecole di grandi dimensioni con carica netta negativa, restano all’interno del neurone.

It all starts with the **squid giant axon**.

Its remarkable size makes it possible to:

- Insert electrodes and perform reliable electrical measurements
- Extract cytoplasm from its interior and measure ionic composition
- Understand the mechanisms of synaptic transmission